

The Honest Harvest

Blockchain-Based Food Supply Chain Tracker

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Introduction



The Honest Harvest Supply Chain

The "Trust Gap"

Normal Databases

They are extremely fast, but they rely on blind trust. Anyone with admin access can secretly change an expiration date or ingredient list.

Pure Blockchains

They are totally unhackable, but saving huge files and long product histories directly on them is way too slow and expensive for real businesses.

Our Solution: The Hybrid Model

Honest Harvest gives you the best of both worlds. We combine the **speed of a normal website** with the **unbreakable security of a blockchain**.

- **Fast:** No waiting for slow networks.
- **Detailed:** Can handle complex tracking.
- **Secure:** Mathematically impossible to fake.



The "Digital Fingerprint"

When you add a product, our system creates a tiny, unique digital fingerprint of all its data (a "Hash").

Digital Fingerprint:

0xac0974bec39a17e36ba4a6b4d238ff94...

We lock **only** this fingerprint on the blockchain. If a hacker alters the database, the fingerprint breaks, exposing the cheat instantly.



Core Architecture



System Overview

Our Ecosystem

We use **Docker** to keep four main tools perfectly in sync:

-  Frontend: React / Next.js
-  Backend: Node.js API
-  Database: PostgreSQL
-  Blockchain: Hardhat



How It Works Behind the Scenes



The Family Tree

Our Postgres database does the heavy lifting, tracking exactly how raw ingredients combine to make final products.



Security First

Our Node.js backend acts like a bouncer, ensuring only verified company managers and employees can make changes.



Low Cost

Our backend only talks to the blockchain when absolutely necessary to lock in a new fingerprint, keeping costs near zero.

The Immutable Ledger

The Smart Contract handles the final step.

Instead of saving massive files, the contract simply records the item ID, the owner, and the tiny digital fingerprint.

This guarantees 100% security and tamper-proofing without slowing down the rest of the application.



Thank You

